

Multiple Choice (10 marks)

1. What is the refracting power of a spherical refracting surface of radius 20 cm separating air ($n = 1$) from glass ($n = 1.5$, where n denotes the index of refraction of the medium)?
 - (a) 1.0 diopter
 - (b) 4.0 diopters
 - (c) 2.0 diopters
 - (d) 2.75 diopters
 - (e) 3.5 diopters
2. The nebular theory of the formation of the solar system successfully predicts all but one of the following. Which one does the theory not predict?
 - (a) The presence of moons around several planets
 - (b) Planets orbit around the Sun in nearly circular orbits in a flattened disk.
 - (c) the equal number of terrestrial and jovian planets
 - (d) The existence of the Sun's magnetic field
 - (e) The different chemical compositions of the planets
3. Under what conditions is the net electric flux through a closed surface proportional to the enclosed charge?
 - (a) only when there are no charges outside the surface
 - (b) only when the charges are in motion
 - (c) only when the electric field is uniform
 - (d) only when the enclosed charge is negative
 - (e) only when the charge is at rest
4. Does an ideal gas get hotter or colder when it expands according to the law $pV^2 = \text{Const}$?
 - (a) remains the same
 - (b) none of these
 - (c) hotter
 - (d) the temperature oscillates
 - (e) it gets colder, but only under low pressure

5. An object is placed 10 cm in front of a diverging mirror. What is the focal length of the mirror if the image appears 2 cm behind the mirror?
- (a) $-2/5$ cm
 - (b) $-3/2$ cm
 - (c) $-2/3$ cm
 - (d) $-5/2$ cm
 - (e) -5 cm

True / False (10 marks)

Answer True or False

- 6. True or False: The thickness of the glass plate that makes an object appear five-sixths of an inch nearer is 1.0 inch.
- 7. True or False: As the angle of an inclined plane increases, the normal force acting on a stationary box remains constant until a critical angle.
- 8. True or False: The diameter of Pluto was accurately measured using brightness measurements during mutual eclipses of Pluto and Charon.
- 9. True or False: The axis of the Earth is tilted at an angle of approximately 23.5 degrees relative to its orbital plane around the Sun.
- 10. True or False: When a siren approaches, its frequency remains constant, while the amplitude decreases as it gets closer to the observer.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. In a double-slit experiment with a slit separation of 0.1 mm and a distance to the screen of 1 m, analyze how the wavelength of yellow light affects the spacing between adjacent bright lines.
- 12. Discuss the energy transition of an electron as it moves from the $n = 3$ to $n = 2$ orbit, and calculate the magnitude of the energy emitted as visible red light during this process.

End of Paper